Class B - General Engineering

Class B - General Engineering

Number of problems: 8
Start at: 09.30 AM
Finish at: 09.53 AM
Duration: 23 Minutes

As an engineer, you need to measure several parameters to determine the phase angle of a single-phase motor. Which parameter should you measure?

$\phi = \tan^{-1}\left(\frac{X_L - X_C}{Z}\right)$

$X_L = 2\pi f.L$

$X_C = \frac{1}{2\pi f.C}$

- ○ No answer
- ○ Voltage, power, current
- ○ Frequency, capacitance, inductance, power, voltage, and current
- ⊗ Frequency, capacitance, inductance

Consider problem shown below.

If $y = f(x)$, find the derivative, $\frac{dy}{dx}$, of $xy^2 + 4x - y\cos(x) = 0$

- ○ Option A: $\frac{xy - y\sin x + 4}{2xy + \cos x}$
- ○ Option B: $\frac{\cos y + 2xy}{x^2 + x\sin y + 4}$
- ○ Option C: $\frac{\cos y - 2xy}{x^2 + x\sin y + 4}$
- ○ Option D: $\frac{y^2 + y\sin x + 4}{2xy + \cos x}$

$2xy.y' + y^2 + 4 - \cos x.y' + y\sin x = 0$

$(\cos x - 2xy)y' = y^2 + y\sin x + 4$

$\frac{dy}{dx} = \frac{y^2 + y\sin x + 4}{\cos x - 2xy}$

The correct combination of the characteristic/eigen value and the characteristic/eigen vector of the matrix

$$\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$\det(A-\lambda I)=0$

$A-\lambda I = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 2-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix}$

$\det(A-\lambda I)$

$= (2-\lambda)(1-\lambda)-0$

$= 2-3\lambda+\lambda^2$

$= \lambda^2-3\lambda+2=0$

$\lambda=2 \quad \lambda=1$

- Option 2: $1, \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $-1, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Option 3: $2, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $-1, \begin{pmatrix} 1 \\ -2 \end{pmatrix}$
- Option 1: $1, \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $0, \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- Option 4: $1, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $2, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Which group of differential equation terms does not match to this ODE:

$$x^3\frac{d^3y}{dx^3} + x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + xy = e^x$$

- First order, third degree, nonlinear
- Third order, first degree, linear
- First degree, third order, nonlinear

Consider the given equation below. If the Newton-Raphson method is used to find root(s) of the equation with guessed/zeroth iteration, xo = 4, what is the third iterated x value, x3?

$x^2 - 4x - 11$

$f'(x) = 2x-4$

$X_{new} = X_{old} - \frac{f(X_{old})}{f'(X_{old})}$

$x_0=4 \rightarrow f(4)=-11$

$f'(4)=4$

$x_1=6,75$

$x_1=6,75 \rightarrow f(6,75)=11,3625$

$f'(6,75)=9,5$

$x_2=5,9539$

$x_2=5,9539 \rightarrow f(5,9539)=0,6333$

$f'(5,9539)=7,9078$

$x_3=5,8738$

- 5.8760
- 5.8790
- 5.8729
- 5.8738

As an engineer, you are required to determine which type of electric motor will be used in a machine. The machine requires a precise control of position and speed. Which type of motor should you used?

- Brushless DC motor
- Stepper motor
- Servo motor
- Brushless AC motor

Consider equation shown below, if x(2) = 10, what is the value of x(3)

$$\frac{dx}{dy} + 3x = 0$$

- 0.034
- 0.059
- 0.014
- 0.049

$\frac{dx}{dy} = -3x$

$\int dy = \int -\frac{dx}{3x}$

$y = -\frac{1}{3}\ln x + C$

$2 = -\frac{1}{3}.\ln 10 + C$

$C = 2{,}768$

$y = -\frac{1}{3}\ln x + 2{,}768$

$\frac{\ln x}{3} = 2{,}768 - y$

$x = e^{3(2{,}768 - y)}$

$x = e^{3(2{,}768 - 3)}$

$x = 0{,}499$

Consider program shown below, if variable M is an integer greater than 0, which equation yields the same meaning with the program?

```
A = X
DO UNTIL M = 0
        Y = A*X
        A = Y
        M = M – 1
END UNTIL
```

- Y=X^M
- Y=X!
- Y=X^(M+1)
- Y=X^(M-1)

Kembali Kirim

Jangan pernah mengirimkan sandi melalui Google Formulir.

Konten ini tidak dibuat atau didukung oleh Google. Laporkan Penyalahgunaan - Persyaratan Layanan - Kebijakan Privasi

Google Formulir